Feature Documentation — Merged 2-Class Model (v16)

soft vs hard

WalkSensePlace

Every feature is computed **per step** (one gait cycle, from one total-pressure peak to the next, ~0.5–1.2 s at $f_s = 62.5\,\mathrm{Hz}$). Sensors used: 12 plantar pressure pads, a 3-axis accelerometer, and a 3-axis gyroscope (the magnetometer is dead and unused). Steps that span a surface change or a `transition`/`wait` tag are excluded, so each step lies on a single surface.

Two conventions appear throughout:

- **"norm" / session-normalised** — divided by a within-recording scale (e.g. that step's mean total pressure) so the value doesn't scale with body weight or how tightly the insole was worn.
- **amplitude-invariant** — the metric is a ratio or shape number, unaffected by overall size.

The target is `soft` (grass / compliant) vs `hard` (dirt, compact ground, concrete, asphalt, exposed aggregate, brick). The model uses **48 features**; the F values are ANOVA soft-vs-hard separation scores (higher = better). Person-confound features that are computed but **dropped** are listed at the end.

Why the soft/hard split is the reliable product. These features primarily encode surface *compliance*. On the soft-vs-hard target the top features separate the classes about twice as sharply as on the 3-class target — e.g. the strongest, `orient_rest_a_roll3_std`, scores $F \approx 3109$ here vs ≈ 1612 in the 3-class model. The features needed to split the *hard* class further (firm natural ground vs pavement) are weak, which is why the 3-class `loose` label doesn't generalise but soft-vs-hard does.

Notation

For one step of length L samples, let p_t be total pressure; $s_{i,t}$ the value of pad i ($i = 1, \dots, 12$); heel sum $h_t = \sum_{i \in \mathcal{H}} s_{i,t}$ and forefoot sum $f_t = \sum_{i \in \mathcal{F}} s_{i,t}$ over the heel set $\mathcal{H}$ and forefoot set $\mathcal{F}$. Accelerometer axes are a_t^x, a_t^y, a_t^z with magnitude $\|a_t\| = \sqrt{(a_t^x)^2 + (a_t^y)^2 + (a_t^z)^2}$; gyroscope $g_t^{\{x,y,z\}}$ analogously. $\bar{x} = \frac{1}{L}\sum_t x_t$ is the mean. $\tilde{p}$ denotes the total-pressure curve resampled to a fixed length and divided by its mean (a unit-scale shape). The *flat-foot instant* is $t^\star = \arg\min_t \|g_t\|$ (foot momentarily still), where the accelerometer points along gravity.

Several features are *step-to-step* quantities: for a per-step scalar $u^{(k)}$ (step index k within a recording+foot sequence), the rolling 3-step standard deviation is

$$\mathrm{roll3\_std}(u)^{(k)} = \sqrt{\tfrac{1}{3}\sum_{j=k-2}^{k}\big(u^{(j)} - \bar{u}^{(k)}\big)^2}, \qquad \bar{u}^{(k)} = \tfrac{1}{3}\sum_{j=k-2}^{k} u^{(j)}. \tag{1}$$

1 Foot-tilt variability (the strongest features)

From a ZUPT-gated orientation estimate: at the flat-foot instant $t^\star$ the foot tilt is read from the accelerometer on two orthogonal axes,

$$\theta_A = \mathrm{atan2}\big(a_{t^\star}^x,\ \sqrt{(a_{t^\star}^y)^2 + (a_{t^\star}^z)^2}\big), \qquad \theta_B = \mathrm{atan2}\big(a_{t^\star}^z,\ \sqrt{(a_{t^\star}^x)^2 + (a_{t^\star}^y)^2}\big). \tag{2}$$

- `orient_rest_a_roll3_std` ($F\approx 3109$) and `orient_rest_b_roll3_std` ($F\approx 1710$) — the **step-to-step variability** of resting foot tilt, i.e. roll3_std(θ_A) and roll3_std(θ_B) via Eq. (**??**). High values mean the foot lands at a different orientation each step — the signature of uneven/compliant ground. The single most class-differentiable features, and they generalise across people. (The *absolute* tilt θ_A, θ_B is a per-person mount offset and is *not* a feature — only its step-to-step variability is kept.)

- `orient_range_a` / `orient_range_b` — the range of foot tilt *within a single stance*, $\max_t \theta_{A,t} - \min_t \theta_{A,t}$: how much the foot rocks while planted.

2 Accelerometer texture / impact (vibration and hardness)

Let x be a step-length signal, detrended, with real-FFT power spectrum $P_k = |X_k|^2$ at frequencies f_k $(k = 1, \ldots, K)$. Define the normalised spectrum $\rho_k = P_k / \sum_j P_j$.

$$\text{spectral centroid} = \frac{\sum_k f_k P_k}{\sum_k P_k}, \qquad \text{HF fraction} = \frac{\sum_{k:\, f_k \geq 5\,\text{Hz}} P_k}{\sum_k P_k}, \tag{3}$$

$$\text{spectral entropy} = \frac{-\sum_k \rho_k \ln \rho_k}{\ln K}, \qquad \text{jerk ratio} = \frac{\text{RMS}(\Delta x)}{\text{RMS}(x - \bar{x})} = \frac{\sqrt{\frac{1}{L-1}\sum_t (x_t - x_{t-1})^2}}{\sqrt{\frac{1}{L}\sum_t (x_t - \bar{x})^2}}. \tag{4}$$

- `accel_jerk_ratio` ($F\approx 2311$) — jerk ratio of the vertical axis a^y (amplitude-invariant); sharp/jerky strikes on hard ground, damped on compliant.

- `accel_hf_frac` ($F\approx 819$) — HF fraction of $\|a_t\|$; higher on firm surfaces.

- `accel_spec_centroid` ($F\approx 693$) — spectral centroid of $\|a_t\|$; shifts up with harder impacts.

- `accel_spec_entropy` — spectral entropy of $\|a_t\|$ (tonal vs broadband texture).

- `accel_zcr` — zero-crossing rate of a^y, $\dfrac{1}{L-1}\sum_t \mathbf{1}\big[\text{sign}(a_t^y - \bar{a}^y) \neq \text{sign}(a_{t-1}^y - \bar{a}^y)\big]$.

- `accel_mag_strike` — peak $\|a_t\|$ in the impact region $\max_{t \leq L/3} \|a_t\|$.

3 Gyroscope motion (foot rotation & stability)

- `gyro_x_std` ($F \approx 1507$) — standard deviation of the x angular rate over the step, $\sigma_{g^x} = \sqrt{\frac{1}{L}\sum_t (g_t^x - \bar{g}^x)^2}$: side-to-side foot rotation/wobble.

- `gyro_x_strike` / `gyro_y_strike` / `gyro_z_strike` — angular rate at the strike instant (rotational "kick" at contact).

- `gyro_y_std` / `gyro_z_std` — angular-rate std on the other two axes.

- `gyro_hf_frac` — HF fraction (same formula) of $\|g_t\|$.

- `gyro_mag_strike` — $\|g_t\|$ at the strike instant.

4 Stance timing & pressure-curve shape

On the unit-scale total curve $\tilde{p}$ (session-normalised shape):

- `stance_duration` ($F \approx 1352$) — loaded time per step, $\#\{t : p_t > \frac{1}{2}\bar{p}\}$; compliant ground softens/lengthens contact.
- `loading_rate` / `loading_rate_norm` — peak rise rate $\max_t \Delta p_t$ (raw and /scale); hard loads faster.
- `rise_slope` / `fall_slope` — pressure rise and fall slopes.
- `slope_asym` — asymmetry between loading and unloading slopes.
- `time_to_peak_frac` — $\arg\max_t p_t / L$ (timing of peak loading).
- `gu_dip_depth` — mid-stance dip depth of $\tilde{p}$, $1 - \dfrac{\min_{t \in \text{mid}} \tilde{p}_t}{\max_t \tilde{p}_t}$; deeper on grass.
- `gu_load_first_frac` — $\sum_{t \le L/2} \tilde{p}_t / \sum_t \tilde{p}_t$.
- `gu_push_peak` — height of the push-off (second) peak of $\tilde{p}$.
- `gu_load_slope` / `gu_unload_slope` — $\max_t \Delta\tilde{p}_t$ (loading) and $\min_t \Delta\tilde{p}_t$ (unloading).

5 Weight distribution (heel vs forefoot)

- `heel_fore_ratio` ($F \approx 665$) — $\bar{h}/\bar{f}$ (front-to-back load).
- `heel_impulse_norm` ($F \approx 695$) — $\sum_t h_t/\text{scale}$ (time-under-load at heel).
- `heel_skew` / `heel_kurt` — skewness and kurtosis of the heel series,

$$\text{skew} = \frac{\frac{1}{L}\sum_t (h_t - \bar{h})^3}{\sigma_h^3}, \qquad \text{kurt} = \frac{\frac{1}{L}\sum_t (h_t - \bar{h})^4}{\sigma_h^4}.$$

- `heel_ripple_frac` — high-frequency “ripple” fraction of the heel signal.
- `midstance_diff` — $h_{t^\star} - f_{t^\star}$ at the flat-foot instant.
- `gu_fore_early_frac` — forefoot loading share in early stance, $\sum_{t \le t_0} \tilde{f}_t / \sum_t \tilde{f}_t$.
- `gu_fore_heel_toff` — $(\arg\max_t \tilde{f}_t - \arg\max_t \tilde{h}_t)/L$.

6 Pressure-map distribution & dynamics (across the 12 pads)

Treat the 12 pads as a spatial map. Let v_i be a per-pad load (e.g. mean over stance, or the flat-foot value) and $q_i = v_i / \sum_j v_j$ its normalised share.

$$\text{spatial entropy} = \frac{-\sum_{i=1}^{12} q_i \ln q_i}{\ln 12}, \qquad \text{participation} = \frac{1}{12} \cdot \frac{1}{\sum_{i=1}^{12} q_i^2}. \tag{5}$$

- `gu_spatial_entropy_stance` ($F \approx 976$) and `gu_spatial_entropy_ff` — normalised Shannon entropy of the pad distribution (whole stance / at flat-foot). High = load spread evenly; low = concentrated. Compliant ground conforms and spreads load.
- `gu_participation_ff` — normalised inverse participation ratio (effective fraction of pads bearing load at flat-foot).
- `gu_active_sensors` — $\#\{i : v_i > 0.1 \max_j v_j\}$ at flat-foot (contact breadth).
- `gu_cop_wander` — mean frame-to-frame change of the normalised map, $\frac{1}{L-1} \sum_t \sum_i |Q_{i,t} - Q_{i,t-1}|$ with $Q_{\cdot,t}$ the per-frame normalised map: how much load shifts around the sole.
- `gu_press_jitter` — mean per-pad temporal jitter, $\mathrm{mean}_i \, \mathrm{std}_t(\Delta^2 s_{i,t})/\mathrm{scale}$.
- `stance_duration_roll3_std` / `loading_rate_roll3_std` / `impact_mag_roll3_std` — step-to-step variability (Eq. (**??**)) of stance duration, loading rate, and impact.

Dropped (computed but excluded from the model)

Removed because cross-user permutation testing showed they act as a **person fingerprint** and *hurt* generalisation to a new person:

- **Per-sensor pressure ratios** — `pressure_01_mean_ratio` ... `pressure_12_mean_ratio`, plus `heel_mean_ratio`, `fore_mean_ratio`, `heel_pmax_ratio`, `fore_pmax_ratio`.
- **Raw magnitudes** — `total_p_mean`, `total_pmax`, `impact_mag`, `impact_mag_norm`, `heel_impulse` (scale with body weight / donning tightness).

How to read the model, in one line

Soft-vs-hard is driven by **step-to-step foot-tilt variability**, **accelerometer jerk/texture**, **gyro rotation variability**, **stance timing**, and **pressure-map spread** — all direct measures of surface *compliance*. Because compliance is exactly what a pressure/IMU insole senses well, this two-class model is the reliable, deployable product.